

**COMPUTER COMPETENCIES REQUIRED FOR E-EXAMINATIONS BY STUDENTS
OF NATIONAL OPEN UNIVERSITY OF NIGERIA IN ENUGU STATE STUDY
CENTRES**

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Abstract

This research is delimited to computer competences required for electronic examinations in Enugu Study Centres of National Open University of Nigeria (NOUN), Enugu State, Nigeria. Two research questions were formulated to guide the study while two null hypotheses were tested. The research was carried out in Enugu State, Nigeria. The population of this study consisted of all the 428 students of National Open University of Nigeria in study centres in Enugu State. The population is made up of 428 students comprising of 309 students in Enugu-Nike road study centre, 41 students of Opi Community Study Centre, 25 students of Awgu Study Centre, and 53 students of Enugu Correctional/ prisons Service Unit. All the students served as sample for the study. A structured questionnaire named "Computer Competencies Required for Electronic Examination Questionnaire (CCREEQ), developed by the researcher was used for data collection. The instrument was validated by two experts. The reliability of the instrument was determined by using Cronbach Alpha Reliability Coefficient to determine the internal consistency of the instrument. The overall reliability coefficient was .75, indicating that the instrument is reliable and suitable for the study. The data collected with the questionnaire were analyzed using Mean ($\bar{x}$) with Standard Deviation (SD) to answer the two research questions. However, each of the null hypotheses were tested using t-test statistics at .05 level of significance. The analyses were done with the use of the Statistical Package for Social Sciences (SPSS). Results of analyses revealed that: the extent to which National Open University of Nigeria students in urban and rural study centres in Enugu State possess computer competencies in word processing and internet research were low. It was recommended, among others, that Educational stakeholders should prioritize the improvement of digital infrastructure in rural study centres and NOUN should develop and implement comprehensive digital literacy training programmes tailored to address the specific needs of students in both urban and rural centres.

Keywords: Computer Competencies, E-Examination, Word Processing and Internet Research, National Open University of Nigeria.

Introduction

The evolution of educational technology began far before 17th century when tribal chiefs and priests systematized bodies of knowledge and early cultures invented pictographs, that is, sign writings to record, preserve, transmit and reproduce information. The primary concern of educational technology is to solve problems that are inherent in education. Ejinwa (2018), opined that the application of modern technology in teaching and learning process started right from the era when B.F. Skinner (1904-1990) experimented with his teaching machine based on the behaviorist theory of stimulus-response-reinforcement which led to the development of programmed instruction. Other pioneers who recognized the potentials of using technology to change the processes of teaching and learning include but are not limited to Seymour Papert (1928-2016), who developed the 'logo' programming language, which is an Educational Programme that enables children to use mathematics as a building material for creating pictures, animations, music, games, and simulations and Patrick Suppes (1922-2014), who is regarded as the father of computer assisted instruction (Roblyer & Doering, 2013). Since then, technology as applied in the field of education has evolved into assorted technological and telecommunication devices and innovative pedagogies enhanced by technology. These marked the beginning of the automation of educational system.

Automation as one of the roles of technology in education gave birth to so many innovations. Presently, the different

technological and communication devices applied in teaching and learning and other educational services done with the help of electronics have blended into the generic term referred to as 'e-learning.' E-learning involves the use of information and communication technologies (ICT) in education (Khitum, 2019). It implies the use of numerous types of technological devices that deliver text, audio, pictures, animation, video and their combinations. Various methods have been devised by educational technologists such as Shraim, and Crompton, (2015), on how best to solve problems of education. Amongst these methods are the use of modern technologies such as computers, handheld devices, audio/video conferencing facilities, internet, CD ROM, tape recorders, television, radio, and currently artificial intelligence. One of the roles of technology in education is automation of educational system. Automation is the use of technology, machinery, or software to perform tasks that were previously done by humans with the aims of increasing efficiency, reducing labour costs and improving accuracy and reliability. Quixy (2023), posited that automation is the process of completing, repeating of mundane tasks using technology which in turn helps one minimize manual works in repetitive admin work and gives one opportunities for quality teaching and learning. For the researcher, the reasons for automation of education and instruction are: to empower knowledge, attract and acquire top talent, make more time, boost productivity and efficiency, streamline communication on one hand and to cater for individual differences in nature

as well as create equal opportunities for everybody to be educated. This of course is the in-thing presently and is so because if everybody is purely educated, the more coherent the world becomes. Equal educational opportunities are one of the core purposes of e-learning.

E-learning is one of the innovations of the automation of educational system. E-learning is highly suitable for Open and Distance Learning Institutions due to its flexibility. The learner decides when, where and how to learn. E-learning integration has brought about a new paradigm shift in education including the development of many innovative technology enhanced learning styles such as Open and Distance Learning (ODL). The various media applied increases attention, concentration, memorization, retention and motivation of the students' learning. It also allows remote teaching and learning as well as collaboration among learners and instructors (Olabode, Fasoranbaku & Oluwadare, 2015). The concept of remote teaching and learning as mentioned above simply denotes the use of technology in distance learning initiatives. According to Williamson, (2018), e-learning is a blanket term used to describe educational initiatives that use electronic means for delivery of educational services. These services may include online learning, computer based learning, web-based learning, collaborative learning, virtual learning, e-Examination among others. These methods and their services are very important in examinations such as in examinations in open and distance learning as they emphasized and are still emphasizing

the directions of education in this 21st century.

Electronic examinations are used in distance learning and the importance cannot be overemphasized. Musingafi, Mapuranga, Chiwanza and Zebron (2015), opined that the use of electronic examinations in open and distance learning provides timely feedback on individual student's achievement. There is however, flaws in implementation in most developing countries. The researcher observed that the developing nations of the world always imitate the developed nations' educational systems without adequate planning for human and non-human resources needed. These types of hasty attitudes toward educational pursuits always create gaps among which is the technical competence in the use of ICT which form one of the challenges confronting electronic examinations in distance learning. The related technical competencies are skills needed to operate ICT hardware and software, skills for networking systems and to access and disseminate information (Ogunleye, 2020). According to Olusegun and Adesoji (2017), ICT competencies is information and communication technology acquired skills that encompasses dexterity, skills, ability and knowledge in computer or web in-depth comprehension and application of basic knowledge on surfing the internet to access and disseminate information. These qualities are highly sought for in this 21st century by everybody particularly for areas like South-Eastern Nigeria that have earned both local and global economic respects. However, inadequate possession of these skills may pose serious digital and human

challenges to productivity in jobs and academic performance irrespective of location. Moreover, enhanced skills and competencies are required to remain at the forefront of the new technological revolution and one of the biggest concerns in terms of technological, social, and cultural advancement in the workplace is the pressure to decrease operation costs and the increased prospect of unemployment faced by people who are less prepared for the changes that are coming (Demirkan, 2016).

The potential of using advanced technology including the elements like machine learning, intelligent process automation, collaborative robotics, decentralized manufacturing management, horizontal and vertical system integration cannot be over emphasized. These will redefine most jobs and at the same time eliminate some of them and generate new ones (Tan, 2017). As the competitiveness of an enterprise is highly dependent on the way work is done, the technology employed, and the know-how of the employees, the competence of the individual is a crucial characteristic (Jensen, 2019). Competence is generally viewed as the main term that encompasses more specific concepts such as knowledge, skills, and abilities among others. One of the main roles of the high schools, universities, as well as adult post-graduation education is to improve these elements, since it is axiomatically aimed at identifying what areas to focus on and the practical orientation of education. This work seeks to determine the extent of possession of computer competencies of National Open University of Nigeria

(NOUN) students in using computers for e-Examinations in Enugu State Study Centres.

The avid thirst for knowledge and provision of equal access to education to every citizen made both the developed and developing nations of the world clamor for Open and Distance Learning (ODL) mode of delivery. Musingafi, et al (2015), attested to accessibility, affordability, flexibility and life based pedagogic opportunities as the benefits derived from learning via Open and Distance Learning strategy. Hence, the introduction of ODL coupled with the rapid growth in information and communication technology in facilitating access and dissemination of knowledge mandated every learner to be ICT compliant in order to enhance competence and improved performance academically and professionally (Ogunlela & Ogunleye, 2014; Ogunleye & Apata, 2018). UNESCO (2014), further reiterated that Open and Distance Learning (ODL) has been translated into a paramount global strategy of providing solutions to the related challenges confronting equal access to education and providing instantaneous feedback to the users via electronic platforms. This position was corroborated by Okopi and Ogunleye (2016), who chronicled the importance of the open and distance education to its recipients and the society at large. This again is a pointer to why every graduate of 21st century should possess computer proficiencies irrespective of their location.

Students of Open and Distance Learning in NOUN need to possess computer competencies and be literate in the

use of computer because e-Examination is a crucial aspect of evaluation in NOUN. Key computer competencies to be possessed by students in NOUN and other universities according to Musingafi, Mapuranga, Chiwanza, and Zebron (2015) include general computer operational principles, basic file management skills, word processing ability, internet research and computer graphics presentation. Therefore, considerable level of electronic competence is required for improved performance to be attained in e-Examinations. NOUN students' computer competency is a significant predictor of improved performance in e-Examination. Osang (2017), asserted that e-examination was intentionally introduced to solve some series of examination variances and challenges encountered like human errors, examination malpractices, technical challenges during compilation of answer scripts which could lead to missing of scripts, leakages of questions as a result of conveying of examination papers to and from the examination venue. Electronic examination is conducted through the internet which affords quick access to students' results after examination. E-Examination is computer-based examination whereby the institution prepared their examination questions to be uploaded to the designated computers that were meant for it. In preparation for e-Examination in National Open University of Nigeria, lecturer in charge or course coordinators are expected to set examination questions and typeset it online using the internet facilities that will guide the students during examination (Shraim, 2018).

Word processing is an application programme that allows one to create letters, reports, newsletters, tables, form-letters, brochures and web pages. This application programme can be used to add pictures, tables and charts to documents. National Council for Educational Research (2018), maintained that the word processing operators achieve considerable time largely due to the faster typing speeds achieved as a result of electronic keyboards and the assistance of word processing software for functions like layout and spell checks among others. Today there are different word processors available; some are proprietary like Microsoft Word, WordPerfect Office, Star Office Writer, and others. Basic functions in all the word processors remain more or less similar and with ease of usage. Shier (2019), described word processing programmes as an invaluable tool for doing assignments and writing essays such that changes can be easily effected. Technottella (2021), provided five proven reasons why word processing should be taught. These reasons include that: word processing is a foundational skill; word processing skills support an efficient workflow; word processing improves quality of projects; word processing skills are transferrable and word processing apps support collaboration. If these important features of word processing are imbibed, students can now effectively carry out internet research.

Internet research is the practice of using Internet information, especially free information on the World Wide Web or Internet-based resources (like Internet Discussion Forum) in research. Internet research has led to a profound input in the

way ideas are formed and knowledge is created. Markham (2021), outlined salient characteristics of the Internet as: The Internet as the medium of communication, in which case it is seen as conduit for transmission of information from one place and person to another; a range of language aids for interacting with others; and a tool for bundling up bits of information into a package that makes sense. The Internet is geographically dispersed, interfacing everywhere, irrespective of location and distance, and enabling the instantaneous and inexpensive transmission of information between people and databases. Internet anonymity makes participants speak freely without restraints brought about by social norms and conventions. The Internet is also multimodal with communication going in multiple modes, alternately or simultaneously among other characteristics. Studies by Yesiloglu (2021), and Amponsa, Aboagye Nanth-kert, Commey-Mintah & Boateng (2022), have attested to the impact of the usage of internet on students' academic performance. It is necessary to note that for a student in distance learning to use other competencies and search the internet effectively, such a student must also be proficient in presentation on graphics.

It then means that a student of distance learning can use these computer competences effectively irrespective of the person's location. Therefore location is considered as an important variable in this study because empirically, no one knows how it affects the acquisition of computer skills.

Location is a person's environments or place in physical space marked with distinguishing features. It is a place one lives and have his being. In the context of this study, location can be only either urban or rural areas. Can location determine the extent of one's acquisition of computer competencies? Location is a factor to reckon with in this study because of the seemingly general belief that those living in urban areas are more enlightened than their counterparts in the rural areas. This assertion had no empirical backing with respect to computer competencies required for electronic examinations. Computer competences required by students for e-Examinations should be similar for both urban and rural students since e-Examinations have no boundaries but the level or extent of acquisitions of it may differ. This gap the study seeks to fill. In this study, those students at Enugu study centre, and those in its affiliate, the Nigeria Correctional Service Study outlet are assumed to be those in urban areas while those at Awgu Community Study Centre together with those at the Opi Community study Centre, Nsukka are regarded as those in rural areas. Generally, it is well known that certain facilities like electricity power houses, network masts, and network of roads and social media among others are more in the cities than in rural areas (Nworgu, 2015). But these cannot be enough reasons to say that those students in the urban areas acquire more computer competencies than their counterparts in the rural areas. This is because individual difference is a factor to reckon with in almost every human endeavor. To this end, it becomes necessary

to factor in location into this study to ascertain empirically the general belief that those in the urban areas acquire more computer competencies than those in the rural areas. The researcher is not aware of a work on computer competencies for NOUN students in Enugu Study Centres within this locality. Therefore, this investigation is necessary and required.

Statement of the Problem

The developing nations of the world such as Nigeria are still backward compared to developed nations in operating their educational systems. The backwardness is centred on lack of adequate preparation in human and material resources. This situation posed great challenges to educational institutions in developing countries. The wastes incurred as a result of the hasty implementations involved are enormous and the products are not only inferior but lack competency compared to their counterparts who schooled in the developed nations. Competency is the taproot of human capital and it is one of the most important sought assets by any business organizations in this 21st century. It has a closer relationships with occupational performance and complex interactions in work environments for productive economy. Competency in any programme is achieved only with proper management of the factors of production. In Nigeria, National Open University of Nigeria with its attendant large population of registered students and considerable number of examinable courses still perform manual processing of scripts. Recently, the university has imbibed e-examination for students in 100 and 200 levels. It is also

worth of note that other human and infrastructural challenges relating to ICT competencies which are required for e-examinations, may not be adequately provided. Research evidence presented in the background of this study attest to the possibility of both facilitators and students of NOUN not possessing key computer competencies of general computer operational principles, file management, word processing, internet research , computer graphics among others. Since e-examination has recently become a key method of assessing students in NOUN, there is need to ascertain extent these students possess these competencies.

No studies known to the researcher had attempted to determine the extent of possession of competencies required by NOUN students for electronic examinations in Enugu State. The dearth of studies in this area seems to cast shadow on the success of e-examinations and assessment of students in Open University as well as in other distance learning programmes in the country. Furthermore, the conflicting effect of variable like location of the study centres on the effective operation of e-exams and subsequently students' competencies in the use of computer had not been adequately addressed. This calls for special attention and investigation in these areas. Therefore, the problem of this study put in question form is to what extent do students possess key computer competencies required for e-examinations in urban and rural study centres in Enugu State?

Purpose of the Study

The purpose of this study was to investigate the extent of possession of computer competencies required for electronic examination by students in Enugu State Study Centres of National Open University of Nigeria. This study therefore determined:

1. The extent to which NOUN students in urban and rural study centres in Enugu State possess computer competencies in Word processing.
2. The extent to which NOUN students in urban and rural study centres of Enugu State possess computer competencies in Internet research.

Research Questions

The following research questions guided this study:

1. To what extent do NOUN students in urban and rural study centres in

Method

The research design for this study is the descriptive survey design. Osuala (2017), asserted that descriptive survey research design involves studying both large and small populations by selecting and studying samples chosen from the population. Okebaram (2014), posited that the main objective of descriptive survey research is to get detailed and factual information about problems and describe them as they are. The study being a descriptive survey research will use a researcher-designed questionnaire to elicit relevant responses from the National Open University of Nigeria (NOUN) students on the computer competencies required for

Enugu state possess computer competencies in Word processing?

2. To what extent do NOUN students in urban and rural study centres in Enugu state possess computer competencies in Internet research?

Hypotheses

The following hypotheses to be tested at 0.05 level of significance guided this study:

1. There is no significant difference between the mean ratings of NOUN students in urban and rural study centres on the extent they possess computer competencies in word processing.
2. There is no significant difference between the mean ratings of NOUN students in urban and rural study centres on the extent they possess computer competencies in internet research.

electronic examinations. The choice for descriptive survey research design as the most suitable for the work was based on the fact that the researcher intends to investigate carefully, drawing out the opinions, attitudes, habits or other social issues directly from the students in the four study centres of the National Open University of Nigeria in Enugu State on the required computer competencies needed for e-examinations. The population of this study will consist of all the 428 students of National Open University of Nigeria in study centres Enugu state. The population is made up of 309 students in Enugu study centre, 41 students in Opi Community study Centre, 25 students in Awgu study centre, and 53 students in Enugu correctional/

prisons service. This population is made up of only 100 and 200 level students of National Open University of Nigeria, Enugu study centres whose mode of examination is purely electronic in nature. The sample for this study consisted of all the 428 students of National Open University of Nigeria, in study centres in Enugu state. No sampling was done because the population was small and manageable. Thus the researcher used all the 428 students of National Open University of Nigeria, study centres in Enugu state for the study.

A structured questionnaire titled “Computer Competencies Required for Electronic Examination in NOUN Questionnaire (CCREEQ), developed by the researcher was used for data collection. The instrument was divided into two sections; A and B. Section A contained the respondents bio data while section B was divided into two clusters with 22 items, structured to provide answers to the research questions that guided the study. Cluster 1 was on Word processing with 11 items and cluster 2 was on Internet research with 11 items. The response format for the instrument was based on a 4-point scale of Very Great Extent (VGE) =4, Great Extent (GE) =3, Low Extent (LE) =2 and Very Low Extent (VLE) =1. The validity of the instrument was ensured as the drafted copies of the instrument together with the research topic, purpose of the study, research questions, hypotheses, and the developed instrument were given to three experts. Two experts in Educational Technology were from the Department of Educational Foundations while one expert in Measurement and Evaluation is of the Department of

Mathematics and Computer Education. All the experts are from the Faculty of Education, Enugu State University of Science and Technology, Enugu. The experts were requested to assess the relevance, adequacy, suitability and comprehensiveness of the items in addressing the research questions as well as the clarity of the instructions given to the respondents. The initial 60 items generated for the study were screened down to 22 items on the questionnaire. The validators’ comments were used to draft the final instrument that was used for data collection.

The reliability of the CCREEQ was determined by administering 30 copies of the questionnaire to a sample of 12 urban and 18 rural National Open University of Nigeria students in Ebonyi State study centre, in order to ascertain the internal consistency of the instrument. The choice for Ebonyi State was due to its nearness to the state of abode of the researcher. The respondents were assured of confidentiality of all the information they supplied. Data collected from the respondents’ responses were analyzed using Cronbach Alpha Reliability Coefficient to determine the internal consistency of the instrument.

The instrument was divided into two sections, each section yielded the following reliability coefficient; cluster 1 had cluster 1 had .79, and cluster 2 had .71, cluster 5 had .98. The overall reliability coefficient was .75, indicating that the instrument is reliable and suitable for the study. The researcher will administer the questionnaire to the respondents with the help of three research assistants. These assistants will be properly briefed on the content of the questionnaire

and its mode of administration. The administration of the questionnaire was done face to face with the researcher and the assistants. Later date was fixed for the collection of questionnaires not filled at the point of administration. The data collected with the questionnaire was analyzed using Mean ($\bar{x}$) with Standard Deviation (SD) to answer the two research questions. However, each of the null hypotheses was tested using t-test statistics at .05 level of significance. The analysis was done with the use of the Statistical Package for Social Sciences (SPSS). For the decision rule; real limits of the mean scores were applied, therefore, the upper and lower limits of the mean were interpreted as follows;

Mean scores from 3.50 – 4.0 (Very Great Extent)

Mean scores from 2.50 – 3.49 (Great Extent)

Mean scores from 1.50 – 2.49 (Low Extent)

Mean scores from 1.49 and below (Very Low Extent)

The null hypotheses will not be rejected when their p -value is equal or less than .05 and will be rejected when the p -value is more than .05 level of significance.

Results

Research Question 1:

To what extent do National Open University of Nigeria students in urban and rural study centres in Enugu states possess computer competencies in word processing?

Table 1: Mean and standard deviation ratings on the extent to which National Open University of Nigeria students in urban and rural study centres in Enugu states possess computer competencies in word processing

n = 416

SN	Items on the: Extent students possess computer competencies in word processing is reflected in their:	Urban		Rural		Overall		Decision
		$\bar{x}$	SD	$\bar{x}$	SD	$\bar{x}$	SD	
1.	Ability to competently use the word processing interface	2.21	.91	2.11	.82	2.19	.90	LE
2.	Have a detailed working understanding of the use of headings	2.19	.92	1.97	.84	2.16	.91	LE
3.	Have a detailed working understanding of the use of paragraphs and lists	2.12	.92	1.95	.82	2.09	.91	LE
4.	Ability to manage long documents that incorporate table of contents	2.17	.91	2.09	.83	2.16	.90	LE
5.	Ability to manage long documents that incorporate table of index entries	2.21	.92	1.91	.87	2.16	.92	LE
6.	Understand sections within a document	2.14	.92	2.05	.82	2.13	.91	LE
7.	Understand sections within a document and can treat sections independently within a single document	2.15	.94	1.86	.77	2.11	.92	LE
8.	Ability to create specific page numbering schemes	2.12	.93	1.86	.83	2.08	.92	LE
9.	Ability to create specific page numbering styles	2.14	.92	2.08	.86	2.13	.91	LE

10.	Have a complete understanding of the application of styles in a document, including the ability to modify preconfigured styles	2.17	.92	2.13	.86	2.16	.91	LE
11.	Have a complete understanding of the application of styles in a document, including the ability to create custom styles	2.14	.91	1.86	.75	2.10	.90	LE

Grand $\bar{x}$ and SD		2.16	0.92	1.99	0.83	2.13	0.91	LE
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Results of analysis in Table 1 showed the mean and standard deviation ratings on the extent to which National Open University of Nigeria students in urban and rural study centers in Enugu states possess computer competencies in Word processing. The results showed that the overall mean responses of items 26 to 36 were less than the cut-off point of 2.50. The overall grand mean responses were low for respondents in

urban ($\bar{x}$ = 2.16, SD = 0.92) and rural ($\bar{x}$ = 1.99, SD = 0.83). The value of the overall mean response ($\bar{x}$ = 2.13, SD = 0.91) was also less than the cut-off point. This implied that the extent to which National Open University of Nigeria students in urban and rural study centres in Enugu states possess computer competencies in word processing was to a Low Extent.

Research Question 2:

To what extent do National Open University of Nigeria students in urban and rural study centres in Enugu state possess computer competencies in internet research?

Table 2: Mean and standard deviation ratings on the extent to which National Open University of Nigeria students in urban and rural study centres in Enugu state possess computer competencies in internet research

n = 416

SN	Items on the: Extent students possess computer competencies in internet research is reflected in their:	Urban		Rural		Overall		Decision
		$\bar{x}$	SD	$\bar{x}$	SD	$\bar{x}$	SD	
12.	Have an appreciation of different kinds of Web search tools: crawler	2.17	.90	1.89	.82	2.13	.89	LE
13.	Have an appreciation of different kinds of Web search tools: directory- based search engines	2.09	.91	2.05	.79	2.08	.89	LE
14.	Have an appreciation of different kinds of Web search tools: meta search engines	2.12	.93	1.98	.77	2.10	.90	LE
15.	Ability to use appropriate search syntax such as Boolean operators	2.13	.92	1.92	.84	2.10	.91	LE
16.	Ability to use appropriate search syntax such as searching on specific phrases	2.15	.93	2.03	.82	2.13	.91	LE
17.	Skills to constrain searches to specific domains	2.11	.93	1.88	.77	2.08	.91	LE
18.	Skills to eliminate or include particular terms from search strings	2.10	.96	2.05	.84	2.09	.94	LE
19.	Have some knowledge of specialized search tools such as paid subscription databases and services	2.13	.91	2.02	.85	2.11	.90	LE
20.	Have some knowledge of specialized search tools such as specialist websites and services	2.16	.93	1.98	.81	2.13	.91	LE

21. Have an understanding of the rigor required for academic research	2.27	.90	2.00	.82	2.23	.89	LE
22. Have a fundamental understanding of reference management tools such as RefWorks	2.14	.91	1.98	.85	2.12	.91	LE
Grand $\bar{x}$ and SD	2.14	0.92	1.98	0.81	2.12	0.91	LE

Results of analysis in Table 2 showed the mean and standard deviation ratings on the extent to which National Open University of Nigeria students in urban and rural study centers in Enugu state possess computer competencies in Internet research. The results showed that the overall mean responses of items 37 to 47 were less than the cut-off point of 2.50. The overall grand mean responses were low for respondents in urban ($\bar{x}$ = 2.14, SD = 0.92) and rural ($\bar{x}$ = 1.98, SD = 0.81). The value of the overall mean response ($\bar{x}$ = 2.12, SD = 0.91) was also less than the cut-off point. This implied

that the extent to which National Open University of Nigeria students in urban and rural study centres in Enugu state possess computer competencies in internet research was to a Low Extent.

Hypothesis 1: There was no significant difference in the mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in word processing

Table 3: t-test of Mean Ratings of Respondents in Urban and Rural study centres on the extent they possess computer competencies in word processing

Location	N	Mean	Std. Deviation	t-cal	df	Sig.	Dec.
Urban	352	2.16	0.54	4.19	414	0.00	S
Rural	64	1.99	0.24				

Table 3 shows that the t-value for the difference in mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in word processing is 4.19 at 0.05 level of significance and 414 degree of freedom. Since the significance value (Sig. =0.00) is

less than 0.05 level of significance, the null hypothesis is rejected as stated. Hence, there is significant difference between the mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in word processing.

Hypothesis 2: There is no significant difference in the mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in internet research.

Table 4: t-test of Mean Ratings of Respondents in Urban and Rural study centres on the extent they possess computer competencies in internet research.

Location	N	Mean	Std. Deviation	t-cal	df	Sig.	Dec.
Urban	352	2.14	0.55	3.92	414	0.00	S
Rural	64	1.98	0.24				

Table 4 shows that the t-value for the difference in mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in internet research is 3.92 at 0.05 level of significance and 414 degree of freedom. Since the significance value (Sig. =0.00) is less than 0.05 level of significance, the null hypothesis is rejected as stated. Hence, there is significant difference between the mean ratings of National Open University of Nigeria students in urban and rural study centres in Enugu state on the extent they possess computer competencies in internet research.

Discussion of Findings

The findings presented in Table 1 indicated that the extent of computer competencies in word processing among National Open University of Nigeria (NOUN) students in both urban and rural study centres in Enugu State was low. Furthermore, the hypothesis testing revealed a significant difference between the mean ratings of students in urban centres compared to those in rural centres, underscoring a disparity in their proficiency in word processing skills. This finding contrasts with the assertions made by Shier (2019), who highlighted word processing programmes as essential tools for completing assignments and writing essays,

emphasizing their ability to facilitate easy modifications and enhance the overall quality of written work. Additionally, Technottella (2021) provided further justification for the importance of teaching word processing skills, noting that proficiency in such programmes is critical for academic success and professional readiness in today's digital landscape. The low competency levels in word processing among NOUN students suggest that many may struggle to effectively utilize these vital tools for their academic tasks, potentially hindering their ability to produce high-quality written assignments. This gap highlights the need for enhanced training programs that focus specifically on developing word processing skills, especially in rural study centres where access to resources and training may be more limited.

The results presented in Table 2 indicated that the extent of computer competencies in internet research among National Open University of Nigeria (NOUN) students in both urban and rural study centres in Enugu State was low. Hypothesis testing further revealed a significant difference between the mean ratings of students in urban centres compared to those in rural centres, highlighting a clear disparity in their ability to conduct effective internet research. This

finding is particularly concerning, as prior studies by Yesiloglu (2021) and Amponsa et al. (2022) have underscored the critical role that internet usage plays in enhancing students' academic performance. The ability to conduct thorough and effective internet research is essential for accessing information, developing critical thinking skills, and engaging with academic content. Therefore, the low levels of competency in this area indicate a pressing need to address this gap.

Educational Implications of the Findings

This study has a number of educational implications for students, facilitators, and other stakeholders.

The findings of this study have important educational implications for students of the National Open University of Nigeria (NOUN). The low levels of computer competencies across both urban and rural study centres, coupled with significant disparities in favour of urban students, highlight a critical need for enhanced digital literacy training and infrastructure development, especially in rural areas. Without sufficient computer skills, students may struggle to engage effectively with NOUN's e-learning, e-examinations and platform, limiting their access to course materials, research tools, and communication channels. To improve learning outcomes, NOUN must prioritize closing the digital divide by providing equal access to computers, reliable internet, and targeted training programmes. This will not only enhance academic performance but also equip students with the necessary skills to thrive in an increasingly digital world.

The findings of this study have important educational implications for facilitators at the National Open University of Nigeria (NOUN). The low computer competencies among students, particularly in rural centres, suggest that facilitators must adopt more proactive approaches to integrating digital skills into their instruction. This includes offering additional support and training in areas like word processing, file management, internet research, and graphics presentation. Facilitators may need to incorporate blended learning strategies, leverage technology to enhance learning experiences, and provide students with more hands-on practice opportunities. Additionally, addressing the significant gap between urban and rural students requires facilitators to tailor their teaching methods to bridge this digital divide, ensuring equitable access to learning resources and fostering the necessary computer literacy for academic and career success.

The findings of this study have significant educational implications for other stakeholders, including policymakers, administrators, and curriculum developers at the National Open University of Nigeria (NOUN). The overall low computer competencies, especially among rural students, highlight the urgent need for investment in digital infrastructure, such as reliable internet access, modern computer labs, and adequate power supply, particularly in rural centres. Policymakers should prioritize policies that bridge the urban-rural digital divide by providing equal access to technological resources. Administrators need to organize regular

digital literacy workshops and ensure that students are adequately supported with the tools they need to succeed. Curriculum developers should revise course materials to integrate more digital skill-building exercises, ensuring that students across all locations develop essential computer competencies critical for modern learning and employment.

Recommendations of the Study

Based on the findings of the study, the researcher recommends as follows:

1. Educational stakeholders should prioritize the improvement of digital infrastructure in rural study centres. This includes ensuring reliable internet access, providing modern computers, and maintaining a consistent power supply. By creating a conducive learning environment with the necessary technological resources, students in rural areas can better develop their computer competencies.
2. NOUN should develop and implement comprehensive digital

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literacy training programmes tailored to address the specific needs of students in both urban and rural centres. These programmes should focus on enhancing skills in general operational principles, file management, word processing, internet research, and graphics presentation. Regular workshops and hands-on training sessions will equip students with the practical skills needed for academic success.

3. Curriculum developers should revise and enhance course materials to incorporate digital skills training as a core component of the learning experience. By integrating technology into the curriculum and emphasizing the importance of computer competencies across all subjects, NOUN can better prepare students for the demands of a technology-driven workforce and promote equitable skill development among all students.

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